

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I Y. Sai Mohith (197372210) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Y. Sai Mohith

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services.
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Evolution from Stand alone Computing to Cloud-based Systems

The evolution of Computing begin with Stand alone Computers, where applications and data were stored and proceed on a single machine. Organisations later adopted client-server over a network. This was followed by distributed Computing where tasks were shared across multiple connected systems to improve performance and reliability.

The emergence of virtualization enabled multiple virtual machines to run on a single physical server, increasing resource utilization. High-Speed internet, broad band connectivity, web services, automation and large-data scale centres further supported the development of cloud Computing.

Key technological transitions:

- 1) High-Speed Internet and broad band
- 2) Server Virtualization
- 3) Distributed and grid Computing
- 4) Web Services and APIs.
- 5) Data Center automation
- 6) High-Capacity Storage Systems

Example: A company once that maintained its own servers can now host application on AWS, Microsoft Azure or Google cloud without purchase hardware.

2) Working of Hypervisors in Virtualization:

A hypervisor is software that creates and manages Virtual Machines (VMs) dividing a physical computer into multiple virtual environments. It allocates CPU, memory, storage and network resources to each VM while keeping them isolated.

Working:

- 1) Hypervision is installed
- 2) Creates multiple VMs.
- 3) Each VM runs its own operating system.
- 4) Resources are allocated dynamically
- 5) Isolation ensures one VM does not effect another.

Diff b/w Type 1 and Type 2 Hypervisor

Type 1

- * Runs directly on hardware
- * Faster and more Secure
- * Used in data centers
- * Better Performance

Ex: VMware, ESXi

Type 2

- * Runs on a host OS.
- * Slightly Slower
- * Used on personal computers
- * Easier to install

Ex: Oracle, Virtual Box

of API's in Cloud Service Integration:

An Application Programming Interface (API) is a set of rules that enables different software applications to communicate.

Role of API's:

- 1) Connected Cloud Services
- 2) Enhance data security
- 3) Automate cloud operations
- 4) Integrate applications from different providers
- 5) Enable third-party application development

How API's Support interoperability:

- 1) Use standard protocols such as REST and HTTP.
- 2) Allow applications built on different platform to communicate
- 3) Enable data sharing between cloud services.
- 4) Support automation and Orchestration.

Example:

A shopping website user payment API's (Razor pay / Stripe) and Google Maps API while storing customer data in AWS cloud.

72) Resource Pooling in Cloud Computing:

Resource pooling is one of the essential characteristics of cloud computing. It means the cloud provider, combines physical resources such as CPU, Memory, Storage and networking into a common pool. These resources are shared among multiple users (called multi-tenancy) and are allocated or de-allocated dynamically based on user demand.

Benefits:

- 1) Efficient utilization of hardware & resources
- 2) Lower infrastructure and maintenance costs.
- 3) Supports multiple users simultaneously
- 4) Provides high availability and better performance
- 5) Enables easy scaling according to demand

Ex. Microsoft Azure and Amazon Web Services (AWS) share their data center resources among thousands of customers.